

LogicPad Pinout and Hardware Configuration

Microcontroller: STM32F103C8 (Cortex-M3)
Project Name: LogicPad (9-Key Macro Keyboard with OLED Display)
This document details the complete pin assignment and interfacing for all external components connected to the STM32F103C8, including the key matrix, RGB multiplexing system, OLED display, and debugging interface.

1. System Core and Debugging

Function	STM32 Pin	Port/Pin	Notes
External Clock (HSE)	OSC_IN	PD0	External 8 MHz Crystal (Atypical for standard blue-pill, but per specification)
External Clock (HSE)	OSC_OUT	PD1	External 8 MHz Crystal (Atypical for standard blue-pill, but per specification)
SWD Debugging Clock	SWCLK	PA14	Serial Wire Debug Clock
SWD Debugging Data	SWDIO	PA13	Serial Wire Debug I/O

2. Key Matrix Pinout (3x3 Array)

The 9 primary keys are connected in a 3x3 matrix, utilizing 3 column outputs and 3 row inputs. The row inputs should be configured with internal Pull-Up resistors.

Matrix Role	Key Role	STM32 Port/Pin	I/O Type
Column 0	C0_OUT	PB5	Output (Active Low/High Drive)
Column 1	C1_OUT	PB4	Output (Active Low/High Drive)
Column 2	C2_OUT	PB3	Output (Active Low/High Drive)
Row 0	R0_IN	PB12	Input (Pull-Up)
Row 1	R1_IN	PB13	Input (Pull-Up)
Row 2	R2_IN	PB14	Input (Pull-Up)

2.1 Dedicated Button

This is a standalone button, separate from the 3x3 matrix.

Key Role	STM32 Port/Pin	Wiring	I/O Type
Dedicated Key	PB15	Connected to GND via the button switch	Input (Pull-Up Required)

3. RGB LED Multiplexing Pinout

This system uses multiplexing (or Charlieplexing logic) to control the color and state of 10 individual RGB LEDs (9 matrix keys + 1 dedicated key). PB8, PB9, and PB10 provide the common color control (R, G, B) for all LEDs, while PA0-PA9 select which individual LED is illuminated. These color control pins should be driven using PWM for intensity control.

LED Function	STM32 Pin	Purpose	Notes
Red Color Control	PB8 (TIM4_CH3)	Common Red Channel	PWM Recommended
Green Color Control	PB9 (TIM4_CH4)	Common Green Channel	PWM Recommended
Blue Color Control	PB10 (TIM2_CH3)	Common Blue Channel	PWM Recommended
COR0 LED Select	PA0	Individual LED Enable / Sink	Key (0, 0)
COR1 LED Select	PA1	Individual LED Enable / Sink	Key (0, 1)
COR2 LED Select	PA2	Individual LED Enable / Sink	Key (0, 2)
C1R0 LED Select	PA3	Individual LED Enable / Sink	Key (1, 0)
C1R1 LED Select	PA4	Individual LED Enable / Sink	Key (1, 1)
C1R2 LED Select	PA5	Individual LED Enable / Sink	Key (1, 2)
C2R0 LED Select	PA6	Individual LED Enable / Sink	Key (2, 0)
C2R1 LED Select	PA7	Individual LED Enable / Sink	Key (2, 1)
C2R2 LED Select	PA8	Individual LED Enable / Sink	Key (2, 2)
Dedicated Key LED Select	PA9 (TIM1_CH2)	Individual LED Enable / Sink	Dedicated Key

4. OLED Display Pinout (I2C)

A 0.96-inch OLED display is connected using the I2C communication protocol.

I2C Function	STM32 Pin	Notes
I2C1 SCL (Clock)	PB6	Requires external Pull-Up resistor
I2C1 SDA (Data)	PB7	Requires external Pull-Up resistor

5. Summary Pinout Table

Port/Pin	Primary Function	Secondary Function
PA0	COR0 LED Select	
PA1	COR1 LED Select	
PA2	COR2 LED Select	
PA3	C1R0 LED Select	
PA4	C1R1 LED Select	
PA5	C1R2 LED Select	
PA6	C2R0 LED Select	
PA7	C2R1 LED Select	
PA8	C2R2 LED Select	
PA9	Dedicated LED Select (TIM1_CH2)	
PA13	SWDIO	Debugging Data
PA14	SWCLK	Debugging Clock
PB3	Column 2 Output	
PB4	Column 1 Output	
PB5	Column 0 Output	
PB6	I2C1 SCL	OLED Clock
PB7	I2C1 SDA	OLED Data
PB8	R_Ctrl (TIM4_CH3)	RGB Common Red (PWM)
PB9	G_Ctrl (TIM4_CH4)	RGB Common Green (PWM)
PB10	B_Ctrl (TIM2_CH3)	RGB Common Blue (PWM)
PB12	Row 0 Input	
PB13	Row 1 Input	
PB14	Row 2 Input	
PB15	Dedicated Button Input	
PD0	External Clock Input (HSE)	<i>Based on user specification</i>
PD1	External Clock Output (HSE)	<i>Based on user specification</i>